



# NORMANHURST BOYS HIGH SCHOOL

## MATHEMATICS EXTENSION 1

2023 Year 12 Course Assessment Task 4 (Trial Examination)

Wednesday August 9th, 2023

### General instructions

- Working time – 2 hours.  
(plus 10 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

### SECTION I

- Mark your answers on the answer grid provided (on page 11)

### SECTION II

- Commence each new question on a new booklet. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: ..... # BOOKLETS USED: .....

Class (please ✓)

☐ 12MAX.1 – Miss C. Kim

☐ 12MXX.1 – Mr Ho

☐ 12MAX.2 – Miss Lee

☐ 12MXX.2 – Mr Sekaran

☐ 12MAX.3 – Miss J. Kim

☐ 12MXX.3 – Ms Ham

Marker's use only.

| QUESTION | 1-10            | 11              | 12              | 13              | 14              | Total           | % |
|----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|---|
| MARKS    | $\overline{10}$ | $\overline{15}$ | $\overline{17}$ | $\overline{17}$ | $\overline{11}$ | $\overline{70}$ |   |

## Section I

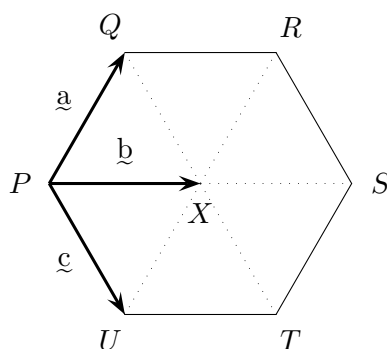
10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer grid provided (labelled as page 11).

- | Questions                                                                                                                                                                                                                                                                           | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1. What is the range of $y = 2 \cos^{-1} x$ ?                                                                                                                                                                                                                                       | 1     |
| (A) $-2 \leq y \leq 2$ (B) $-\pi \leq y \leq \pi$ (C) $0 \leq y \leq \pi$ (D) $0 \leq y \leq 2\pi$                                                                                                                                                                                  |       |
| 2. It is given that $f(x) = e^{x+2}$ . Which of the following is the inverse function $f^{-1}(x)$ ?                                                                                                                                                                                 | 1     |
| (A) $f^{-1}(x) = -2 + \log_e x$ (C) $f^{-1}(x) = e^{x-2}$                                                                                                                                                                                                                           |       |
| (B) $f^{-1}(x) = 2 + \log_e x$ (D) $f^{-1}(x) = e^{x+2}$                                                                                                                                                                                                                            |       |
| 3. How many numbers greater than 2100 can be formed with the digits 1, 2, 3, 4 if no digit is to be used more than once?                                                                                                                                                            | 1     |
| (A) 16      (B) 18      (C) 24      (D) 48                                                                                                                                                                                                                                          |       |
| 4. How many solutions does the equation $x^{\frac{1}{3}} =  x - 2  - 3$ have?                                                                                                                                                                                                       | 1     |
| (A) 0      (B) 1      (C) 2      (D) 3                                                                                                                                                                                                                                              |       |
| 5. $PQRSTU$ is a regular hexagon with side lengths of 4 cm and is divided into six equilateral triangles. It is given that $\overrightarrow{PQ} = \underline{a}$ , $\overrightarrow{PX} = \underline{b}$ and $\overrightarrow{PU} = \underline{c}$ , as shown in the diagram below. | 1     |



Which of the following is the value of  $\underline{a} \cdot (\underline{a} + \underline{b} + \underline{c})$ ?

- (A) 8      (B) 16      (C) 32      (D) 48

6. Which of the following expressions is equal to  $\sin x - 3 \cos x$ ? 1

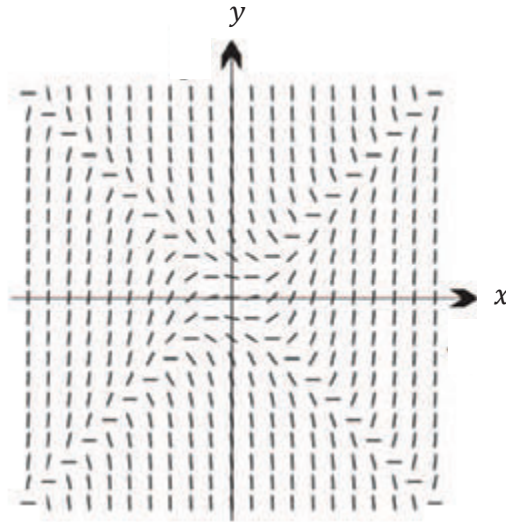
(A)  $\sqrt{10} \cos(x + \tan^{-1} 3)$

(C)  $\sqrt{10} \cos\left(x + \tan^{-1} \frac{1}{3}\right)$

(B)  $\sqrt{10} \sin(x - \tan^{-1} 3)$

(D)  $\sqrt{10} \sin\left(x - \tan^{-1} \frac{1}{3}\right)$

7. Which of the following differential equations best corresponds to the slope field shown below? 1



(A)  $y' = \frac{1}{4}(-x^2 + y^2)$

(C)  $y' = \frac{1}{4}(x^2 - y^2)$

(B)  $y' = -\frac{1}{4}(x^2 + y^2)$

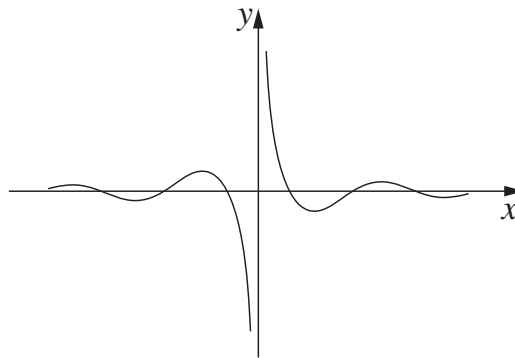
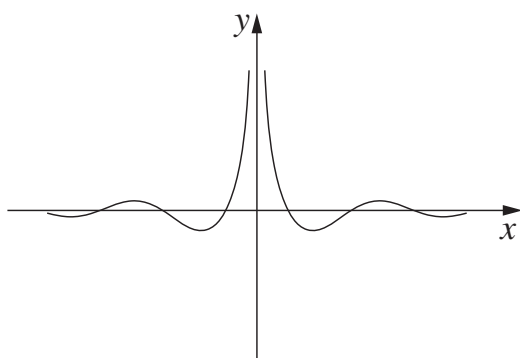
(D)  $y' = \frac{1}{4}(x^2 + y^2)$

**Examination continues overleaf...**

8. Which diagram best represents the graph  $y = \frac{\cos x}{x}$ ? 1

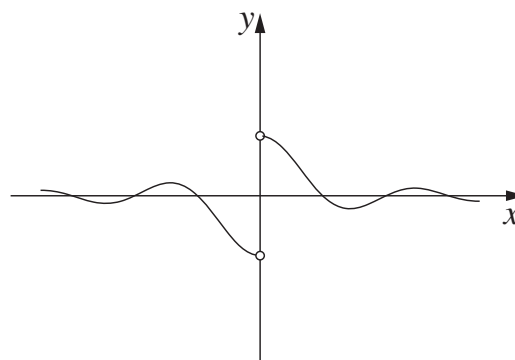
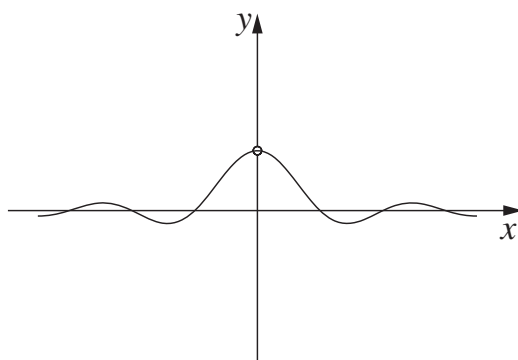
(A)

(C)



(B)

(D)



9. A particle is moving in a straight line such that its displacement,  $x$ , from the origin after  $t$  seconds is given by  $x = -3 \cos^2 t$ . 1

Which of the following best describes the motion of the particle when  $t = \frac{5\pi}{6}$

- (A) The particle is moving in the positive direction with an increasing speed  
 (B) The particle is moving in the positive direction with a decreasing speed  
 (C) The particle is moving in the negative direction with an increasing speed  
 (D) The particle is moving in the negative direction with a decreasing speed

10. Consider the expansion of 1

$$(1 + x + x^2 + \cdots + x^n) (1 + 2x + 3x^2 + \cdots + (n+1)x^n)$$

What is the coefficient of  $x^n$  when  $n = 50$ ?

- (A) 1225                      (B) 1300                      (C) 1326                      (D) 1431

## Section II

60 marks

Attempt Questions 11 to 14

Allow approximately 1 hours and 45 minutes for this section.

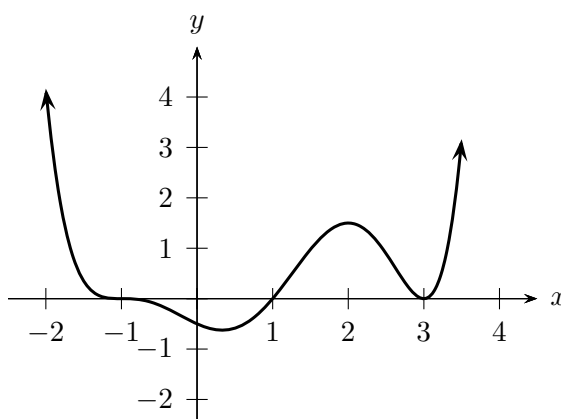
Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

**Question 11** (15 Marks)

Commence a NEW booklet.

**Marks**

- (a) If  $f(x) = \tan^{-1}\left(\frac{x}{2}\right)$ , find the value of  $f'(2)$ . **2**
- (b) It is given that  $\cos 2A = \frac{2}{5}$ . Find the value of  $\sin A \sin 5A + \cos A \cos 5A$ . **2**
- (c) The graph of the polynomial function  $y = P(x)$  is given below.



- i. Without using calculus, write down a possible equation of  $y = P(x)$ . **1**
- ii. Solve  $P(x) \leq 0$  for  $x$ . **2**
- (d) i. By considering  $(1+x)^{2n}$ , or otherwise, show that **2**

$$\sum_{k=0}^{2n} \binom{2n}{k} \left(-\frac{1}{2}\right)^k = \left(\frac{1}{2}\right)^{2n}$$

- ii. Hence evaluate

$$\sum_{k=0}^{2n-1} \binom{2n}{k} \left(-\frac{1}{2}\right)^k$$

**2**

- (e) There are twelve people that need to be divided into groups.
- i. In how many ways can they be divided into two groups of six people? **1**
- ii. In how many ways can they be divided into two groups of six people if two particular people must be in the same group? **1**
- iii. In how many ways can they be seated around one circular table, if the two groups of six people from (i) must be seated among their own groups? **2**

**Examination continues overleaf...**

**Question 12** (17 Marks)

Commence a NEW booklet.

**Marks**

- (a) Consider the polynomial  $P(x) = ax^7 + bx^6 + 1$ , where  $a, b \in \mathbb{R}$  and  $a, b \neq 0$ . Find the values of  $a$  and  $b$  when  $P(x)$  is divisible by  $(x - 1)^2$ . **3**

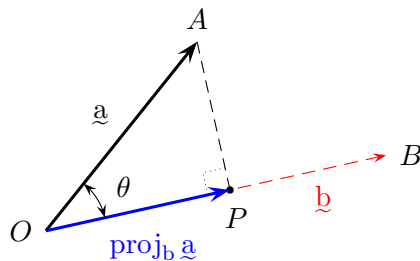
- (b) It is given that  $t = \tan A$  and  $p \cos 2A - \sin 2A = 1$ , where  $p \in \mathbb{R}$ . Show that **3**

$$t = \frac{p-1}{p+1}$$

where  $0 \leq A \leq \frac{\pi}{2}$ .

- (c) By using the substitution  $x = \frac{1}{4}(u^2 + 5)$ , find  $\int 3x\sqrt{4x-5} \, dx$ . **3**

- (d) The diagram below shows  $\overrightarrow{OA} = \underline{\mathbf{a}}$  and  $\overrightarrow{OB} = \underline{\mathbf{b}}$  and  $\theta$  is the acute angle between  $\overrightarrow{OA}$  and  $\overrightarrow{OB}$ .



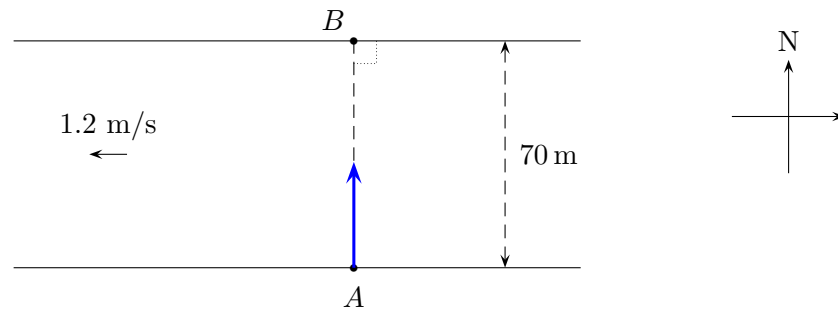
- i. By using the fact that  $\underline{\mathbf{a}} \cdot \underline{\mathbf{b}} = |\underline{\mathbf{a}}| |\underline{\mathbf{b}}| \cos \theta$ , or otherwise, prove that **2**

$$\text{proj}_{\underline{\mathbf{b}}} \underline{\mathbf{a}} = \frac{\underline{\mathbf{a}} \cdot \underline{\mathbf{b}}}{\underline{\mathbf{b}} \cdot \underline{\mathbf{b}}} \underline{\mathbf{b}}$$

It is given that  $\underline{\mathbf{a}} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$  and  $\underline{\mathbf{b}} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$ .

- ii. Find a perpendicular vector to  $\underline{\mathbf{b}}$ . **1**
- iii. Hence, or otherwise, find the shortest distance from the point  $A$  to the line  $OB$  by using vector methods. **2**

- (e) A fish at point  $A$  wants to swim across the river that is 70 m wide. The banks of the river are parallel and the points  $A$  and  $B$  are on opposite sides of the river. **3**



The fish swims at a speed of 1.8 m/s and the river is flowing downstream at 1.2 m/s. However, a 0.5 m/s wind blows constantly to the north-westerly direction.

What is the bearing that the fish should head to in order to land directly at point  $B$  from point  $A$ ?

**Examination continues overleaf...**

**Question 13** (17 Marks)

Commence a NEW booklet.

**Marks**

- (a) Prove by mathematical induction that for all positive integer values of  $n$ , **3**

$$(1^2 \times 2^1) + (2^2 \times 2^2) + (3^2 \times 2^3) + \cdots + (n^2 \times 2^n) = (n^2 - 2n + 3) \times 2^{n+1} - 6$$

- (b) The temperature  $T_1$  of a beaker of chemical, and the temperature  $T_2$  of a surrounding barrel of cooler water, satisfy the following equations:

$$\frac{dT_1}{dt} = -k(T_1 - T_2)$$

$$\frac{dT_2}{dt} = \frac{3}{4}k(T_1 - T_2)$$

where  $k$  is a positive constant.

- i. Show, by differentiation, that  $\frac{3}{4}T_1 + T_2 = C$ , where  $C$  is a constant. **1**
- ii. Hence, show that  $\frac{dT_1}{dt} = kC - \frac{7}{4}kT_1$ . **1**
- iii. By integration, show that  $T_1 = \frac{4}{7}C + Be^{-\frac{7}{4}kt}$ , where  $B$  is a constant, is a **3**  
solution to the differential equation  $\frac{dT_1}{dt}$  in (ii).

Initially, the beaker of chemical had a temperature of  $T_1 = 120^\circ\text{C}$  and the barrel of water had a temperature of  $T_2 = 22^\circ\text{C}$ . Ten minutes later, the temperature of the beaker of chemical had fallen to  $90^\circ\text{C}$ .

- iv. Show that **2**  
$$T_1 = 64 + 56e^{-\frac{7}{4}kt}$$
- v. Show that eventually the beaker of chemical and the surrounding barrel of **2**  
water reach the same temperature.

Mr Kim found out that the number of molecules  $N$  in the beaker of chemical is inversely proportional to the temperature of the beaker,  $T_1$ , which is given by  $N = \frac{1000}{T_1}$ .

- vi. Show that the rate of growth of  $N$  is **3**

$$\frac{dN}{dt} = \frac{7k}{4}N \left( 1 - \frac{64}{1000}N \right)$$

- vii. Hence, or otherwise, find the maximum rate of growth of  $N$  given that **2**  
 $k = 0.001$ .

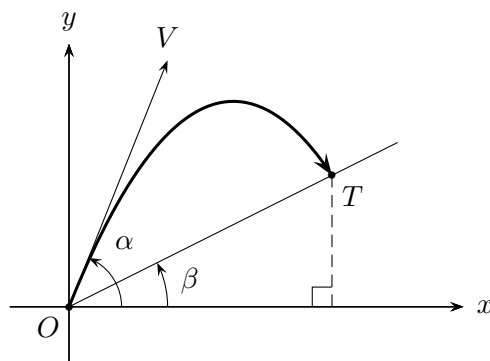
**Question 14** (11 Marks)

Commence a NEW booklet.

**Marks**

- (a) The diagram shows an inclined plane that makes an angle of  $\beta$  with the horizontal. A projectile is fired from  $O$ , at the bottom of the incline, with a speed of  $V$  m/s at an angle of elevation  $\alpha$  to the horizontal, as shown below. **3**

The projectile hits the inclined plane at  $T$ .



The displacement vector,  $t$  seconds after projection, is given by

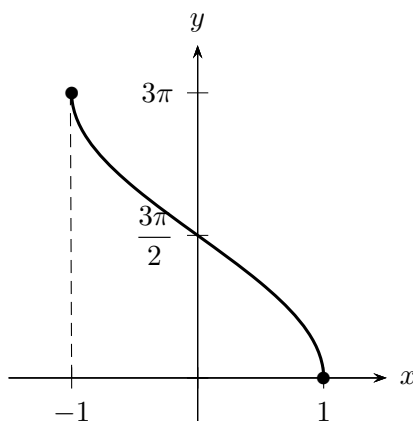
$$\underline{r} = (Vt \cos \alpha) \underline{i} + (-5t^2 + Vt \sin \alpha) \underline{j}$$

(Do NOT prove this.)

Show that the time taken to hit the inclined plane is

$$t = \frac{V}{5}(\sin \alpha - \cos \alpha \tan \beta)$$

- (b) Below is the graph of the function  $y = 3 \cos^{-1} x$ . **4**



Find the exact volume of the solid of revolution formed if the area bounded by  $y = 3 \cos^{-1} x$ , the coordinate axes and the line  $x = -1$  is rotated about the  $y$ -axis.

**Examination continues overleaf...**

- (c) Consider the following functions:

4

$$y = \sin x \cos x$$

$$y = mx$$

where  $m < 0$ .

It is given that there is only one point of intersection at  $x = x_0$  between the two functions for  $x > 0$ .

Show that there exists another value  $x = x_1$ , such that  $x_1 \in \left(\pi, \frac{3\pi}{2}\right)$  and

$$m = \cos x_1 \quad \text{and} \quad \tan x_1 = x_1$$

(*Hint*: Draw picture)

**End of paper.**

## Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

NESA STUDENT #: .....

Class (please ✓)

☐ 12MXX.1 – Ms C.Kim

☐ 12MXX.1 – Mr Ho

☐ 12MXX.2 – Ms Lee

☐ 12MXX.2 – Mr Sekaran

☐ 12MAX.3 – Mr Lam/ Ms J.Kim

☐ 12MXX.3 – Ms Ham

### Directions for multiple choice answers

- Read each question and its suggested answers.
- Select the alternative (A), (B), (C), or (D) that best answers the question.
- Mark only one circle per question. There is only *one* correct choice per question.
- Fill in the response circle completely, using blue or black pen, e.g.

(A) (B) ● (D)

- If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) (B) ~~●~~ ●

- If you continue to change your mind, write the word **correct** and clearly indicate your final choice with an arrow as shown below:

(A) (B) ~~●~~ ~~●~~ <sup>correct</sup>

1 – (A) (B) (C) (D)

6 – (A) (B) (C) (D)

2 – (A) (B) (C) (D)

7 – (A) (B) (C) (D)

3 – (A) (B) (C) (D)

8 – (A) (B) (C) (D)

4 – (A) (B) (C) (D)

9 – (A) (B) (C) (D)

5 – (A) (B) (C) (D)

10 – (A) (B) (C) (D)

## Sample Band 6 Responses

### Section I

1. (D) 2. (A) 3. (B) 4. (C) 5. (B)  
6. (B) 7. (C) 8. (C) 9. (D) 10. (C)

### Section II

#### Question 11

- (a) (2 marks)

- ✓ [1] for  $f'(x)$   
✓ [1] for final answer

$$f'(x) = \frac{2}{x^2 + 4}$$
$$f'(2) = \frac{2}{8} = \frac{1}{4}$$

- (b) (2 marks)

- ✓ [1] for  $\cos 4A$   
✓ [1] for final answer

$$\begin{aligned}\sin A \sin 5A + \cos A \cos 5A &= \cos 5A \cos A + \sin 5A + \sin A \\ &= \cos(5A - A) \\ &= \cos 4A \\ &= 2 \cos^2 2A - 1 \\ &= 2 \left(\frac{2}{5}\right)^2 - 1 \\ &= -\frac{17}{25}\end{aligned}$$

- (c) i. (1 mark)

$$P(x) = (x + 1)^3(x - 1)(x - 3)^2$$

- ii. (2 marks)

- ✓ [1] for  $-1 \leq x \leq 1$   
✓ [1] for final answer

$$-1 \leq x \leq 1 \text{ or } x = 3$$

- (d) i. (2 marks)

- ✓ [1] for expanding  $(1 + x)^{2n}$   
✓ [1] for final result

$$(1+x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} x^k$$

Let  $x = -\frac{1}{2}$

$$\begin{aligned} \sum_{k=0}^{2n} \binom{2n}{k} \left(-\frac{1}{2}\right)^k &= \left(1 - \frac{1}{2}\right)^{2n} \\ &= \left(\frac{1}{2}\right)^{2n} \end{aligned}$$

ii. (1 mark)

$$\begin{aligned} \sum_{k=0}^{2n-1} \binom{2n}{k} \left(-\frac{1}{2}\right)^k &= \sum_{k=0}^{2n} \binom{2n}{k} \left(-\frac{1}{2}\right)^k - \binom{2n}{2n} \left(-\frac{1}{2}\right)^{2n} \\ &= \left(\frac{1}{2}\right)^{2n} - \left(\frac{1}{4}\right)^n \\ &= 0 \end{aligned}$$

(e) i. (1 mark)

$$\frac{{}^{12}C_6 \times {}^6C_6}{2} = 462 \text{ ways}$$

ii. (1 mark)

$${}^{10}C_4 \times {}^6C_6 = 210 \text{ ways}$$

iii. (2 marks)

$$6! \times 6! = 518400 \text{ ways}$$

### Question 12

(a) (3 marks)

- ✓ [1] for  $P(1) = P'(1) = 0$
- ✓ [1] for (1) and (2)
- ✓ [1] for final answer

$$\begin{aligned} P(x) &= ax^7 + bx^6 + 1 \\ P'(x) &= 7ax^6 + 6bx^5 \end{aligned}$$

$\therefore x = 1$  is a double root,  $P(1) = P'(1) = 0$ .

$$\begin{aligned} a + b + 1 &= 0 \quad \dots (1) \\ 7a + 6b &= 0 \quad \dots (2) \end{aligned}$$

$$(2) - (1) \times 6,$$

$$a = 6 \text{ and } b = -7$$

(b) (3 marks)

- ✓ [1] for substituting into  $\cos 2A$  and  $\sin 2A$
- ✓ [1] for making substantial progress
- ✓ [1] for final result

If  $t = \tan A$ ,  $\cos 2A = \frac{1-t^2}{1+t^2}$  and  $\sin 2A = \frac{2t}{1+t^2}$ .

$$\begin{aligned}
 p \cos 2A - \sin 2A &= 1 \\
 p \times \frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2} &= 1 \\
 p(1-t^2) - 2t &= 1+t^2 \\
 p(1-t^2) &= t^2 + 2t + 1 \\
 p(1-t)(1+t) &= (t+1)^2 \\
 p(1-t) &= 1+t \quad \dots \left( \because \tan A > 0 \text{ for } 0 \leq A \leq \frac{\pi}{2}, 1+t \neq 0 \right) \\
 t(1+p) &= p-1 \\
 \therefore t &= \frac{p-1}{p+1}
 \end{aligned}$$

(c) (3 marks)

- ✓ [1] for  $dx = \frac{u}{2} du$
- ✓ [1] for substitution and integration
- ✓ [1] for final answer

$$\begin{aligned}
 x &= \frac{1}{4}(u^2 + 5), \quad u^2 = 4x - 5, \quad u = \sqrt{4x - 5} \\
 dx &= \frac{1}{4} \times 2u \, du = \frac{u}{2} \, du \\
 \therefore \int 3x\sqrt{4x-5} \, dx &= \frac{3}{4} \int (u^2 + 5)\sqrt{u^2} \times \frac{u}{2} \, du \\
 &= \frac{3}{8} \int u^4 + 5u^2 \, du \\
 &= \frac{3}{8} \left( \frac{u^5}{5} + 5 \times \frac{u^3}{3} \right) + c \\
 &= \frac{3}{40} (\sqrt{4x-5})^5 + \frac{5}{8} (\sqrt{4x-5})^3 + c \\
 &= \frac{3}{40} (4x-5)^{\frac{5}{2}} + \frac{5}{8} (4x-5)^{\frac{3}{2}} + c
 \end{aligned}$$

(d) i. (2 marks)

- ✓ [1] for (\*)
- ✓ [1] for final result

$$\text{In } \triangle AOP, \quad \cos \theta = \frac{OP}{|\underline{a}|}$$

$$\begin{aligned} \text{proj}_{\underline{b}} \underline{a} &= OP \hat{\underline{b}} \quad \dots (*) \\ &= |\underline{a}| \cos \theta \hat{\underline{b}} \end{aligned}$$

$$\therefore \cos \theta = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| |\underline{b}|},$$

$$\begin{aligned} \text{proj}_{\underline{b}} \underline{a} &= \cancel{|\underline{a}|} \frac{\underline{a} \cdot \underline{b}}{\cancel{|\underline{a}|} |\underline{b}|} \hat{\underline{b}} \\ &= \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|} \frac{\underline{b}}{|\underline{b}|} \\ &= \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|^2} \underline{b} \\ &= \frac{\underline{a} \cdot \underline{b}}{\underline{b} \cdot \underline{b}} \underline{b} \end{aligned}$$

ii. (1 mark)

$$\begin{pmatrix} -6 \\ 3 \end{pmatrix} \text{ or } \begin{pmatrix} -2 \\ 1 \end{pmatrix} \text{ or any other vector that is perpendicular}$$

iii. (2 marks)

✓ [1] for  $\text{proj}_{\underline{a}} \underline{u}$

✓ [1] for final answer

$$\text{Let } \underline{u} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}.$$

$$\begin{aligned} \text{proj}_{\underline{a}} \underline{u} &= \frac{\begin{pmatrix} -2 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \end{pmatrix}}{\sqrt{(-2)^2 + 1^2}} \hat{\underline{u}} \\ &= \frac{4 + 5}{\sqrt{5}} \hat{\underline{u}} \\ &= \frac{9}{\sqrt{5}} \hat{\underline{u}} \end{aligned}$$

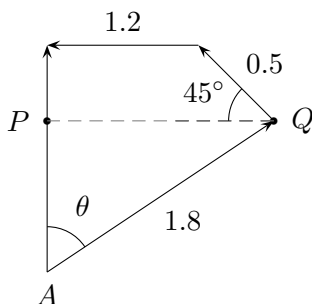
$$\therefore \text{The shortest distance from } A \text{ to the line } OB \text{ is } \left| \text{proj}_{\underline{a}} \underline{u} \right| = \frac{9}{\sqrt{5}}.$$

(e) (3 marks)

✓ [1] for vector diagram or correct vertical vector

✓ [1] for  $PQ$ , or equivalent merit

✓ [1] for final answer



$$PQ = 1.2 + \frac{0.5}{\sqrt{2}} = \frac{6}{5} + \frac{1}{2\sqrt{2}}$$

In  $\triangle APQ$ ,

$$\begin{aligned}\sin \theta &= \frac{\frac{6}{5} + \frac{1}{2\sqrt{2}}}{1.8} \\ \theta &= 59.6647\dots^\circ\end{aligned}$$

$\therefore$  The bearing that the fish should head to is  $060^\circ$  T (to the nearest degree)

### Question 13

(a) (3 marks)

- ✓ [1] for proving the base case
- ✓ [1] for use of the assumption
- ✓ [1] for final result

Let  $P(n)$  be the given proposition.

- Prove  $P(1)$  is true

$$\text{LHS} = 1^2 \times 2^1 = 2$$

$$\text{RHS} = (1^2 - 2 + 3) \times 2^2 - 6 = 2$$

$\therefore$  LHS = RHS,  $P(n)$  is true for  $n = 1$ .

- Assume  $P(k)$  is true

$$(1^2 \times 2^1) + (2^2 \times 2^2) + (3^2 \times 2^3) + \dots + (k^2 \times 2^k) = (k^2 - 2k + 3) \times 2^{k+1} - 6$$

- Prove  $P(k+1)$  is true

$$\begin{aligned}(1^2 \times 2^1) + (2^2 \times 2^2) + \dots + ((k+1)^2 \times 2^{k+1}) &= ((k+1)^2 - 2(k+1) + 3) \times 2^{k+2} - 6 \\ &= (k^2 + 2k + 1 - 2k - 2 + 3) \times 2^{k+2} - 6 \\ &= (k^2 + 2) \times 2^{k+2} - 6\end{aligned}$$

$$\begin{aligned}
\text{LHS} &= (k^2 - 2k + 3) \times 2^{k+1} - 6 + (k+1)^2 \times 2^{k+1} \quad \dots \text{from the assumption} \\
&= 2^{k+1}(k^2 - 2k + 3 + k^2 + 2k + 1) - 6 \\
&= 2^{k+1} \times 2(k^2 + 2) - 6 \\
&= (k^2 + 2) \times 2^{k+2} - 6 \\
&= \text{RHS}
\end{aligned}$$

$\therefore P(k+1)$  is true.

• Conclusion

By mathematical induction, the proposition is true for all positive integers.

(b) i. (1 mark)

$$\begin{aligned}
\frac{d}{dt} \left( \frac{3}{4}T_1 + T_2 \right) &= \frac{3}{4} \frac{dT_1}{dt} + \frac{dT_2}{dt} \\
&= -\frac{3}{4}k(T_1 - T_2) + \frac{3}{4}k(T_1 - T_2) \\
&= 0
\end{aligned}$$

$\therefore \frac{3}{4}T_1 + T_2 = C$ , where  $C$  is a constant.

ii. (1 mark)

$$\therefore T_2 = C - \frac{3}{4}T_1,$$

$$\frac{dT_1}{dt} = -k \left( T_1 - \left( C - \frac{3}{4}T_1 \right) \right) = -k \left( T_1 - C + \frac{3}{4}T_1 \right) = -k \left( \frac{7}{4}T_1 - C \right) = kC - \frac{7}{4}kT_1$$

iii. (3 marks)

- ✓ [1] for separation of variables and integration
- ✓ [1] substantial progress in making  $T_1$  the subject
- ✓ [1] for final result

$$\begin{aligned}
\frac{dT_1}{dt} &= kC - \frac{7}{4}kT_1 \\
\int \frac{1}{C - \frac{7}{4}T_1} dT_1 &= \int k dt \\
-\frac{4}{7} \ln \left| C - \frac{7}{4}T_1 \right| &= kt + A \quad \dots \text{where } A \text{ is a constant} \\
\ln \left| C - \frac{7}{4}T_1 \right| &= -\frac{7}{4}kt + A_1 \\
C - \frac{7}{4}T_1 &= \pm e^{-\frac{7}{4}kt + A_1} \\
&= A_2 e^{-\frac{7}{4}kt} \\
\therefore \frac{7}{4}T_1 &= C - A_2 e^{-\frac{7}{4}kt} \\
T_1 &= \frac{4}{7}C - A_3 e^{-\frac{7}{4}kt} \quad \dots \text{where } A_1, A_2, A_3 \text{ are constants} \\
\therefore T_1 &= \frac{4}{7}C + B e^{-\frac{7}{4}kt} \quad \dots \text{where } B \text{ is a constant}
\end{aligned}$$

iv. (2 marks)

✓ [1] for substitution of  $t = 0$ ,  $T_1 = 120$  or the value of  $C$

✓ [1] for  $B$

When  $t = 0$ ,  $T_1 = 120^\circ\text{C}$  and  $T_2 = 22^\circ\text{C}$

$$C = \frac{3}{4}T_1 + T_2 = \frac{3}{4} \times 120 + 22 = 112$$

and

$$T_1 = \frac{4}{7} \times 112 + B e^{-\frac{7}{4}kt}$$

$$120 = 64 + B$$

$$\therefore B = 56$$

$$\therefore T_1 = 64 + 56e^{-\frac{7}{4}kt}$$

v. (2 marks)

✓ [1] for showing  $T_1 \rightarrow 64$

✓ [1] for showing  $T_2 \rightarrow 64$

As  $t \rightarrow \infty$ ,  $e^{-\frac{7}{4}kt} \rightarrow 0$

$$\therefore T_1 \rightarrow 64$$

$$\therefore T_2 = C - \frac{3}{4}T_1,$$

As  $T_1 \rightarrow 64$ ,

$$\begin{aligned} T_2 &\rightarrow 112 - \frac{3}{4} \times 64 \\ &= 64 \end{aligned}$$

$\therefore$  The beaker and the surrounding barrel eventually reach the same temperature.

vi. (3 marks)

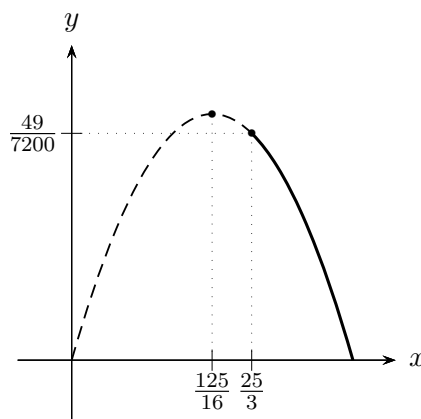
✓ [1] for  $\frac{dN}{dt}$

✓ [1] for substantial progress in manipulation of  $\frac{dN}{dt}$

✓ [1] for final result

$$\begin{aligned}
 N &= \frac{1000}{64 + 56e^{-\frac{7}{4}kt}} \\
 \frac{dN}{dt} &= -\frac{1000}{\left(64 + 56e^{-\frac{7}{4}kt}\right)^2} \times \left(-\frac{7}{4}k \times 56e^{-\frac{7}{4}kt}\right) \\
 &= \frac{1000}{64 + 56e^{-\frac{7}{4}kt}} \times \frac{7}{4}k \times \frac{56e^{-\frac{7}{4}kt}}{64 + 56e^{-\frac{7}{4}kt}} \\
 &= N \times \frac{7k}{4} \times \frac{64 + 56e^{-\frac{7}{4}kt} - 64}{64 + 56e^{-\frac{7}{4}kt}} \\
 &= N \times \frac{7k}{4} \times \left(1 - \frac{64}{64 + 56e^{-\frac{7}{4}kt}}\right) \\
 &= \frac{7k}{4}N \left(1 - \frac{64}{1000}N\right)
 \end{aligned}$$

vii. (2 marks)



$\therefore$  Initially when  $T_1 = 120$ ,  $N = \frac{1000}{120} = \frac{25}{3}$  and  $N \geq \frac{25}{3}$ , thus as shown in the graph above,  $\max \frac{dN}{dt}$  is when  $N = \frac{25}{3}$ .

$\therefore \max \frac{dN}{dt} = \frac{7}{4} \times 0.001 \times \frac{25}{3} \left(1 - \frac{64}{1000} \times \frac{25}{3}\right) = \frac{49}{7200} \approx 0.0068$  molecule/min (to 2 d.p.)

### Question 14

(a) (3 marks)

- ✓ [1] for the coordinates of  $T$ , or equivalent merit
- ✓ [1] for equating  $\underline{r}$  with  $T$ , or equivalent merit
- ✓ [1] for final result

$$\underline{r} = \begin{pmatrix} (V \cos \alpha) t \\ -5t^2 + (V \sin \alpha) t \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$\therefore T(OT \cos \beta, OT \sin \beta)$  and equating with  $\underline{r}$ ,

$$(V \cos \alpha) t = OT \cos \beta \quad \dots (1)$$

$$-5t^2 + (V \sin \alpha) t = OT \sin \beta \quad \dots (2)$$

$$(2) \div (1),$$

$$\begin{aligned}\tan \beta &= \frac{-5t^2 + (V \sin \alpha) t}{(V \cos \alpha) t} \\ &= \frac{-5t + (V \sin \alpha)}{(V \cos \alpha)} \\ \therefore -5t &= V \cos \alpha \tan \beta - V \sin \alpha \\ t &= -\frac{V}{5} (\cos \alpha \tan \beta - \sin \alpha) \\ &= \frac{V}{5} (\sin \alpha - \cos \alpha \tan \beta)\end{aligned}$$

(b) (4 marks)

✓ [1] for  $V_{\text{cylinder}}$

✓ [1] for integrating the expression of  $V_1$

✓ [1] for evaluating  $V_1$

✓ [1] for final answer

$$\cos^{-1} x = \frac{y}{3}, x^2 = \cos^2 \left( \frac{y}{3} \right)$$

$$\begin{aligned}V &= V_{\text{cylinder}} - V_1 \\ V_{\text{cylinder}} &= \pi(1)^2 \times 3\pi = 3\pi^2 \\ V_1 &= \pi \int_{\frac{3\pi}{2}}^{3\pi} \cos^2 \left( \frac{y}{3} \right) dy \\ &= \frac{\pi}{2} \int_{\frac{3\pi}{2}}^{3\pi} 1 + \cos \frac{2y}{3} dy \\ &= \frac{\pi}{2} \left[ y + \frac{3}{2} \sin \frac{2y}{3} \right]_{\frac{3\pi}{2}}^{3\pi} \\ &= \frac{\pi}{2} \left( 3\pi + \frac{3}{2} \sin 2\pi - \frac{3\pi}{2} - \frac{3}{2} \sin \pi \right) \\ &= \frac{\pi}{2} \times \frac{3\pi}{2} \\ &= \frac{3\pi^2}{4} \\ \therefore V &= 3\pi^2 - \frac{3\pi^2}{4} = \frac{9\pi^2}{4}\end{aligned}$$

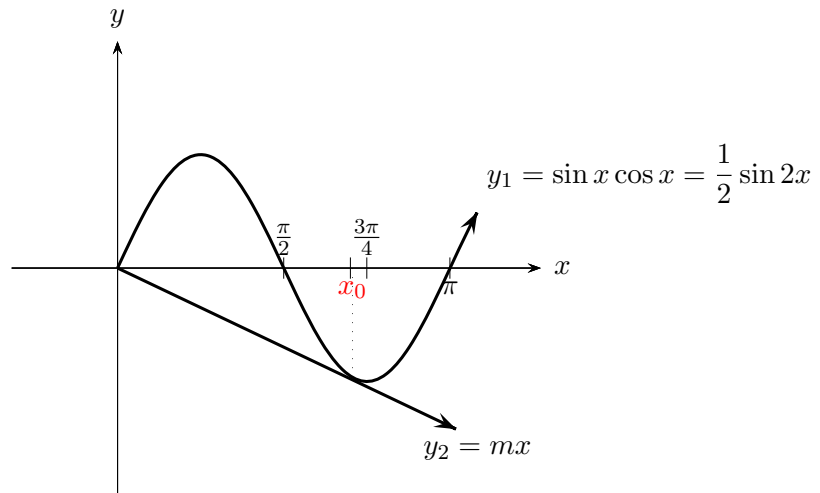
(c) (4 marks)

✓ [1] for the two graphs shown

✓ [1] for showing  $m = \cos(2x_0)$

✓ [1] for showing  $\tan x_1 = x_1$

✓ [1] for showing  $x_1 \in \left( \pi, \frac{3\pi}{2} \right)$



$$\frac{dy_1}{dx} = \cos 2x$$

$$\frac{dy_2}{dx} = m$$

$$\therefore \text{At } x = x_0, m = \cos(2x_0)$$

$$\text{Let } x_1 = 2x_0,$$

$$m = \cos x_1 \quad \dots (1)$$

Also since their y-coordinates are equal,

$$\begin{aligned} \frac{1}{2} \sin(2x_0) &= mx_0 \\ \sin 2x_0 &= m 2x_0 \\ \sin x_1 &= mx_1 \\ \therefore x_1 &= \frac{\sin x_1}{m} \\ &= \frac{\sin x_1}{\cos x_1} \quad \dots \text{from (1)} \\ &= \tan x_1 \end{aligned}$$

$$\text{And from the graph, } \frac{\pi}{2} < x_0 < \frac{3\pi}{4}, \pi < 2x_0 < \frac{3\pi}{2}$$

$$\therefore x_1 \in \left( \pi, \frac{3\pi}{2} \right)$$